



CHULA ENGINEERING
Foundation toward Innovation

COMPUTER

DS Module Summary

2110531 Data Science and Data Engineering Tools (2026/1)

Prof. Peerapon Vateekul, Ph.D.
Department of Computer Engineering,
Faculty of Engineering, Chulalongkorn University
Peerapon.v@chula.ac.th
www.cp.eng.chula.ac.th/~peerapon/

+ Lots of Topics: DS/DE/Viz

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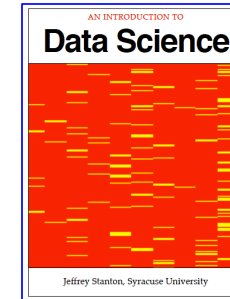
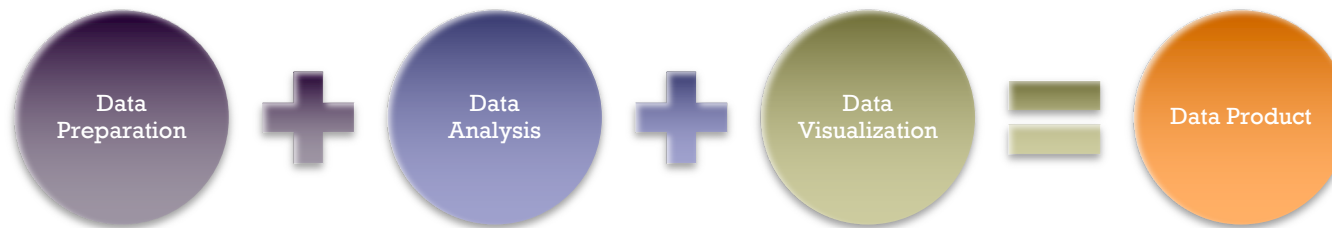
#	Tue (1PM-4PM)	Sat (1PM-4PM)	Topics	Professor	Module
1	04-Aug-26	08-Aug-26	Introduction, Pandas, Data Prep + AI-Assisted	Aj.Peerapon	DS1
2	11-Aug-26	15-Aug-26	Traditional ML (1)	Aj.Peerapon	DS2
3	18-Aug-26	22-Aug-26	Traditional ML (2)	Aj.Peerapon	DS3
4	25-Aug-26	29-Aug-26	Deep Learning (1); CNN, RNN (LSTM, GRU)	Aj.Peerapon	DS4
5	01-Sep-26	05-Sep-26	Deep Learning (2); Transformer Model monitoring (MLflow) Hyperparameter Tuning (Optuna)	Aj.Peerapon	DS5
6	08-Sep-26	12-Sep-26	Advanced topics (Prompt Engineering, RAG, Agentic AI); Tools: LangChain, LangGraph	Aj.Peerapon	DS6
7	15-Sep-26	19-Sep-26	Guest speaker + Midterm Announcement	Aj.Peerapon	DS7
	22-Sep-26	26-Sep-26	Midterm Exam Week (21 - 25 Sep 2026)		
8	29-Sep-26	03-Oct-26	Data visualization (Graduation Week 29 Sep - 1 Oct)	Aj.Veera	VIZ1
9	06-Oct-26	10-Oct-26	Data Storage (Data Lakehouse; Open Table Format) + Vector Database	Aj.Natawut	DE1
10	13-Oct-26	17-Oct-26	Web Scraping (LLM) + Data ingestion (Kafka) Note: 13-Oct-26 is King Bhumibol Adulyadej the Great Memorial Day	Aj.Natawut	DE2
11	20-Oct-26	24-Oct-26	Big data processing (Spark)	Aj.Natawut	DE3
12	27-Oct-26	31-Oct-26	MLOps: Orchestration (Airflow) and serving (FastAPI, Seldon Core)	Aj.Natawut	DE4
13	03-Nov-26	07-Nov-26	Spatial analysis	Aj.Veera	VIZ2
14	10-Nov-26	14-Nov-26	Graph analysis AI-Assited Data Analytics	Aj.Veera	VIZ3
15	17-Nov-26	21-Nov-26	MCP, AI Integration, Harness Engineering, Evaluation, Guardrails, Observability	Aj.Natawut	DE5
	24-Nov-26	28-Nov-26	Final Exam Week (23 Nov - 4 Dec 2026) *** Sat 28 Nov 2026 at 1PM-4PM ***		

- Module1 Assignment (DS) 10%
- Module2 Assignment (DE) 10%
- Module3 Assignment (Viz) 5%
- Midterm Exam (Kaggle) 20%
- Project 20%
- Attendance 5%
- Final Exam 30% (Lab Test)



What is Data Science (DS)? (aka. data analytics)

- Data
 - Facts and statistics collected for reference or analysis
- Science
 - A systematic study through observation and experiment
- Data Science
 - The scientific exploration of data to extract meaning or insight,
 - and the construction of software to utilize such insight in a business context.

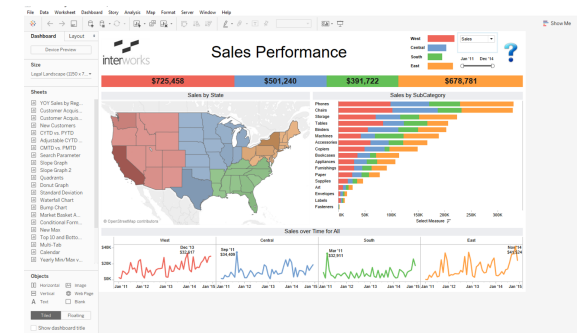
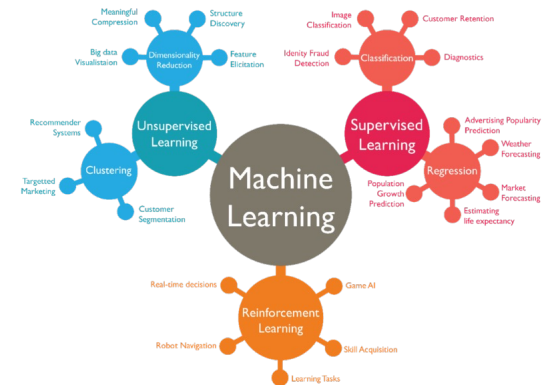
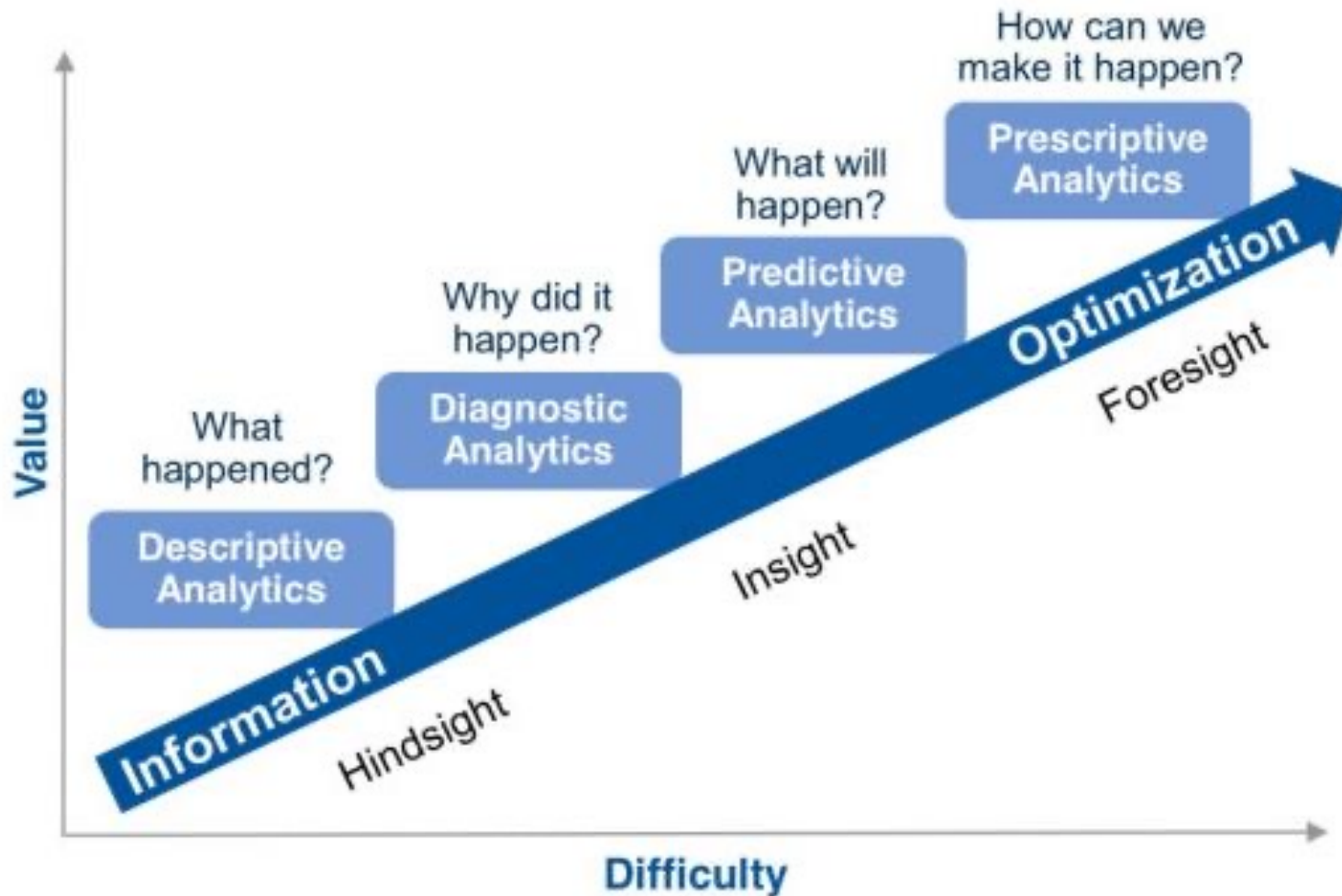


1. Transform data into **valuable insights**
2. Transform data into **data products**
3. Transform data into **interesting stories**



Ta Virod Chiraphadhanakul
Data Scientist, Facebook

Data Analytics (Data Science)





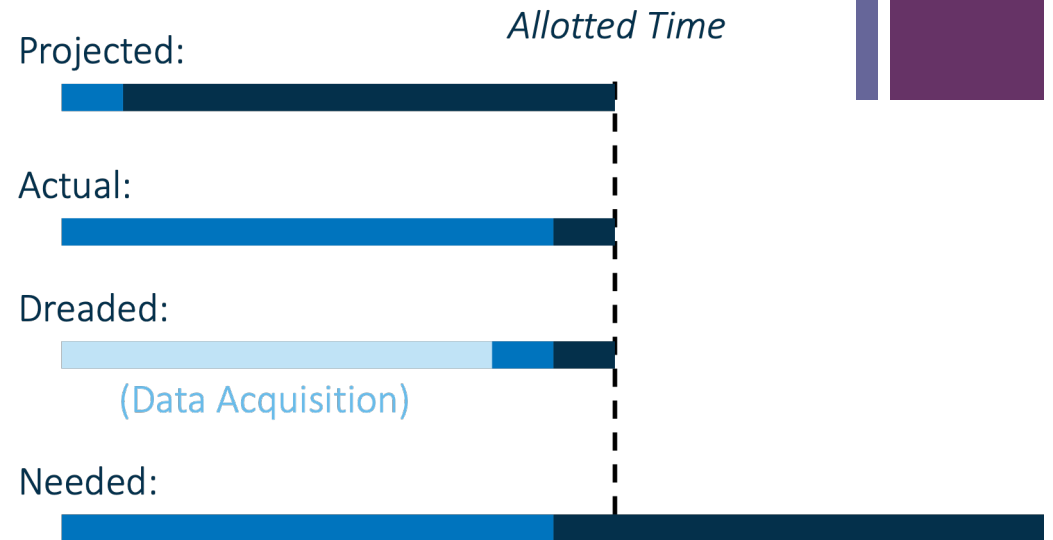
Data is the most important factor. The first thing you should present is **not the model**, but the data.

IN



=

OUT



Data Preparation

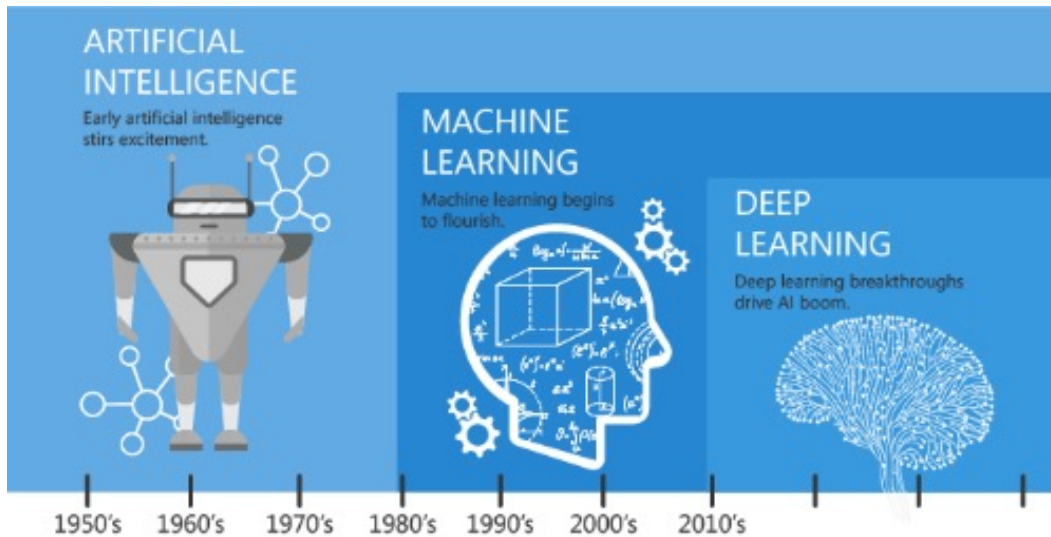
Data Analysis



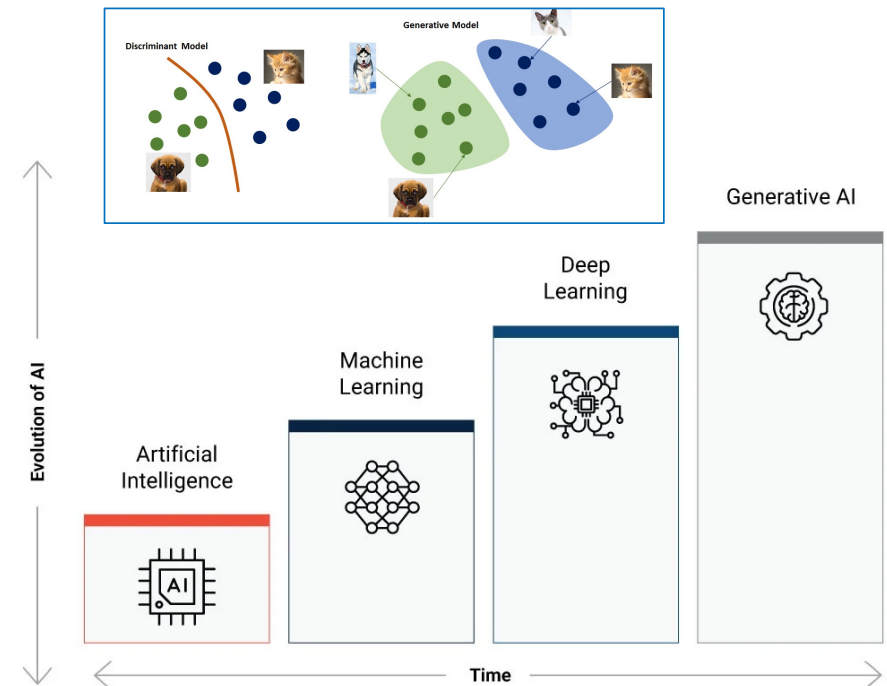
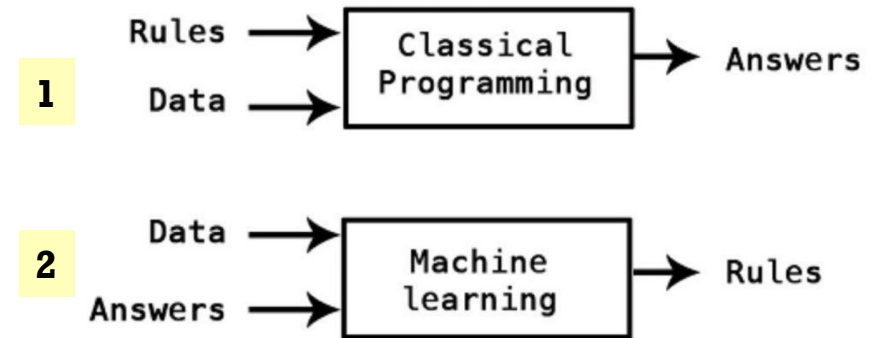


AI = Automation

- 1) Rule-based AI
- 2) Machine Learning (ML)



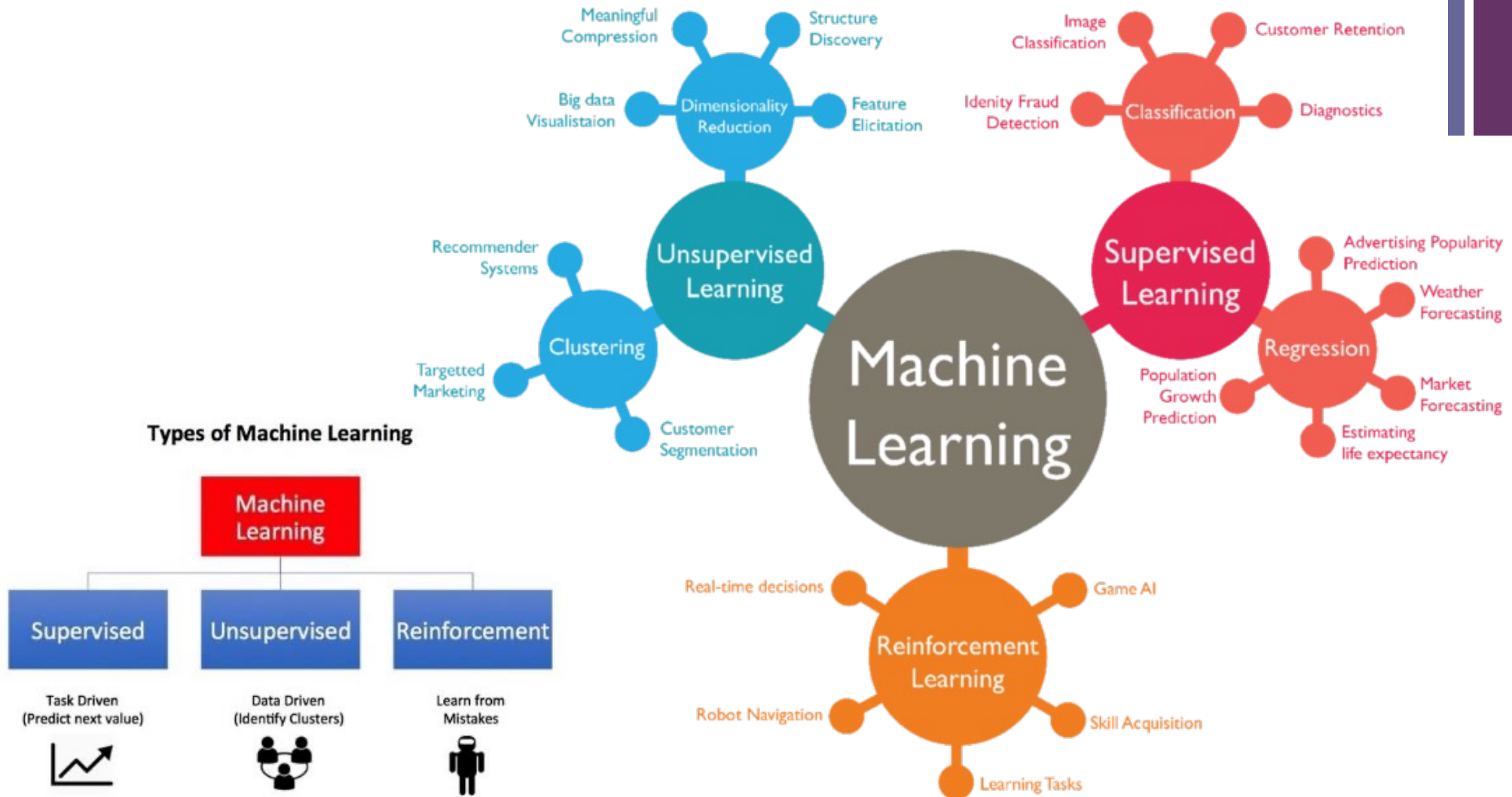
Since an early flush of optimism in the 1950's, smaller subsets of artificial intelligence - first machine learning, then deep learning, a subset of machine learning - have created ever larger disruptions.



<https://mc.ai/machine-learning-basics-artificial-intelligence-machine-learning-and-deep-learning/>

+ Machine Learning (cont.)

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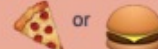




Prediction Algorithms

- Decision Tree
- (Logistic) Regression
- Neural Networks (NN)
- kNN
- Support Vector Machine
- Deep Learning

BASIC REGRESSION

- LINEAR** `linear_model.LinearRegression()`
Lots of numerical data 
- LOGISTIC** `linear_model.LogisticRegression()`
Target variable is categorical 

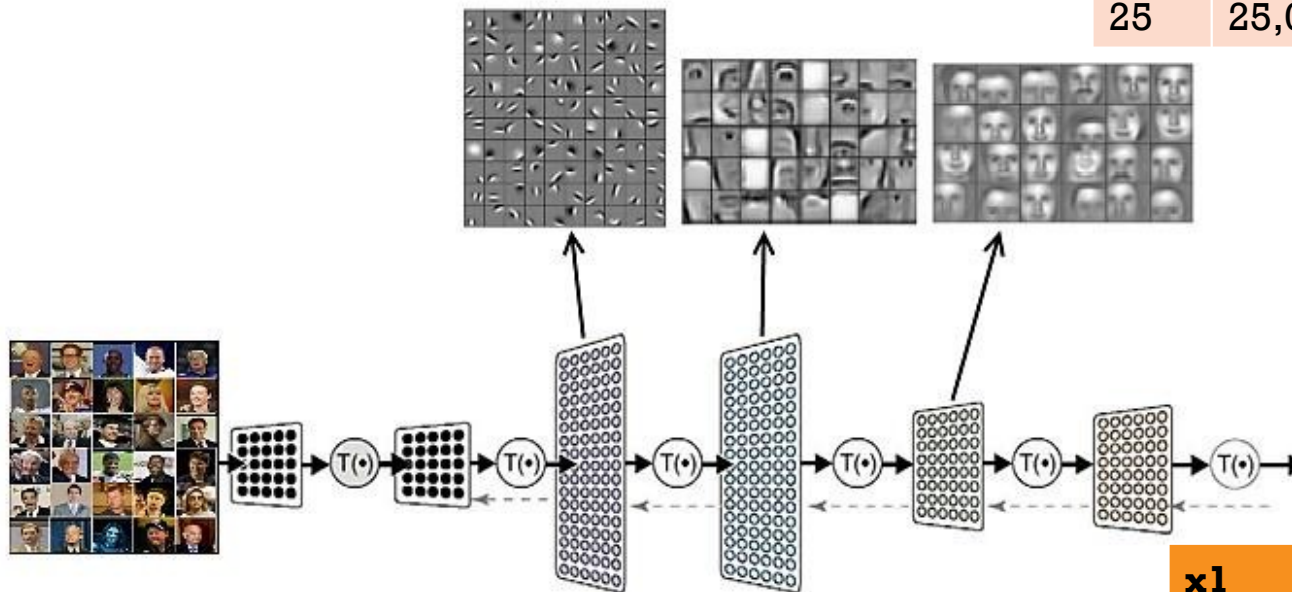
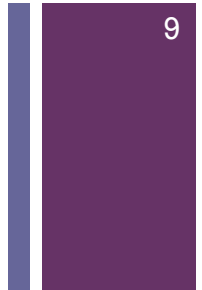
CLASSIFICATION

- NEURAL NET** `neural_network.MLPClassifier()`
Complex relationships. Prone to overfitting
Basically magic. 
- K-NN** `neighbors.KNeighborsClassifier()`
Group membership based on proximity 
- DECISION TREE** `tree.DecisionTreeClassifier()`
If/then/else. Non-contiguous data
Can also be regression 
- RANDOM FOREST** `ensemble.RandomForestClassifier()`
Find best split randomly
Can also be regression 
- SVM** `svm.SVC()` `svm.LinearSVC()`
Maximum margin classifier. Fundamental
Data Science algorithm 
- NAIVE BAYES** `GaussianNB()` `MultinomialNB()` `BernoulliNB()`
Updating knowledge step by step with new info 

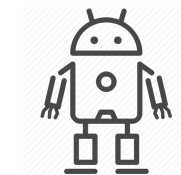


Deep Learning

A deep neural network consists of a **hierarchy of layers**, whereby each layer **transforms the input data** into more abstract representations (e.g., edge -> nose -> face). The output layer combines those features to make predictions.



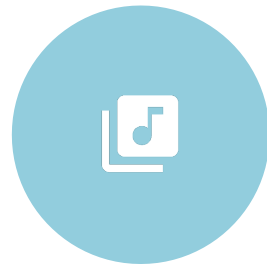
Age	Income	Gender	Province	Corona
25	25,000	Female	Bangkok	Yes



x1	x2	x3	x4	Corona
0.7	0.2	-0.5	-0.1	Yes



Deep Learning Application



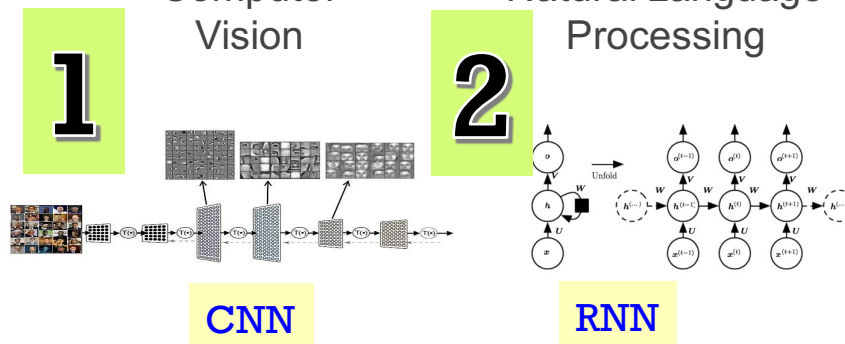
Speech
Recognition



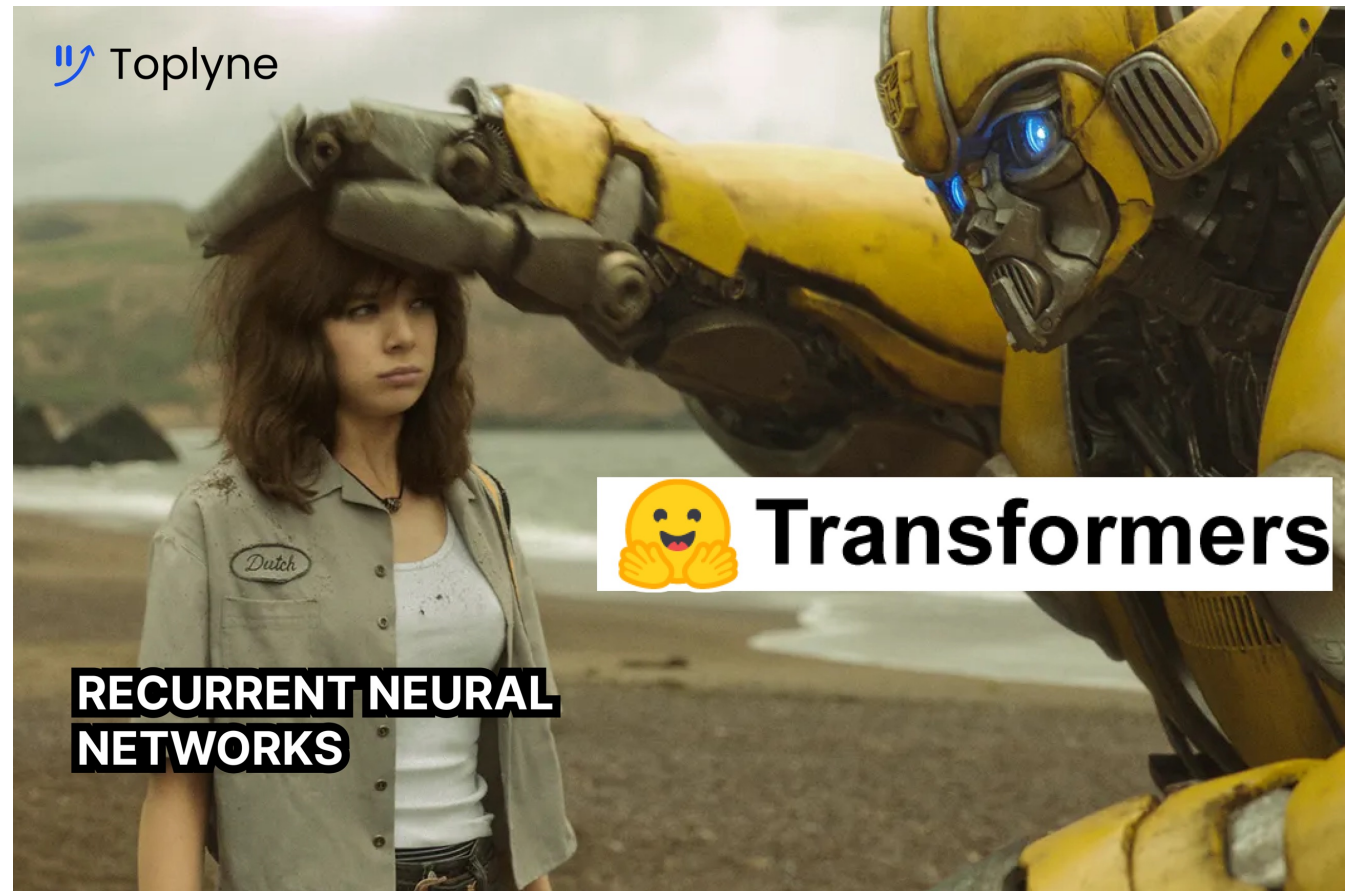
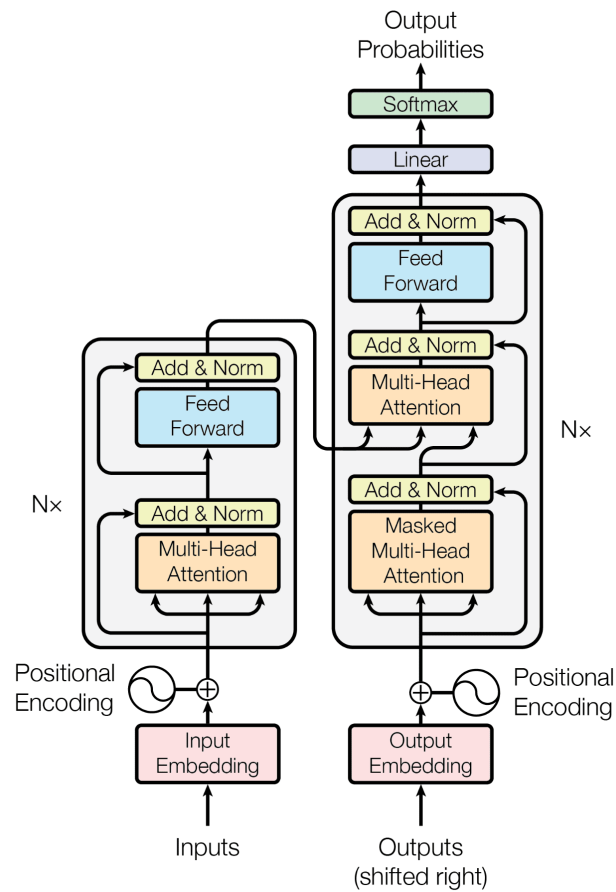
Computer
Vision



Natural Language
Processing



Rise of Transformer (2017)



Large + LM = LLM

"Large" is no longer just about parameter count.

Model	Year	Parameters	Key advancement
GPT-1	2018	117M	Transformer language modeling
GPT-2	2019	1.5B	Stronger zero-shot generation
GPT-3	2020	175B	In-context & few-shot learning
GPT-4	2023	ND (~1.7TB)	Multimodality + stronger reasoning
GPT-5	2025	ND (~300B)	Integrated reasoning + tool use
GPT-6 Astra	2026	Not disclosed (ND)	Computer use + end-to-end agentic work

Modern LLM capability ≠ parameter count alone

Pretraining + Architecture + Post-training +
Inference-time Reasoning + Tools

Evolution of LLMs

From language models to agentic models



GPT-6 Astra (2026)

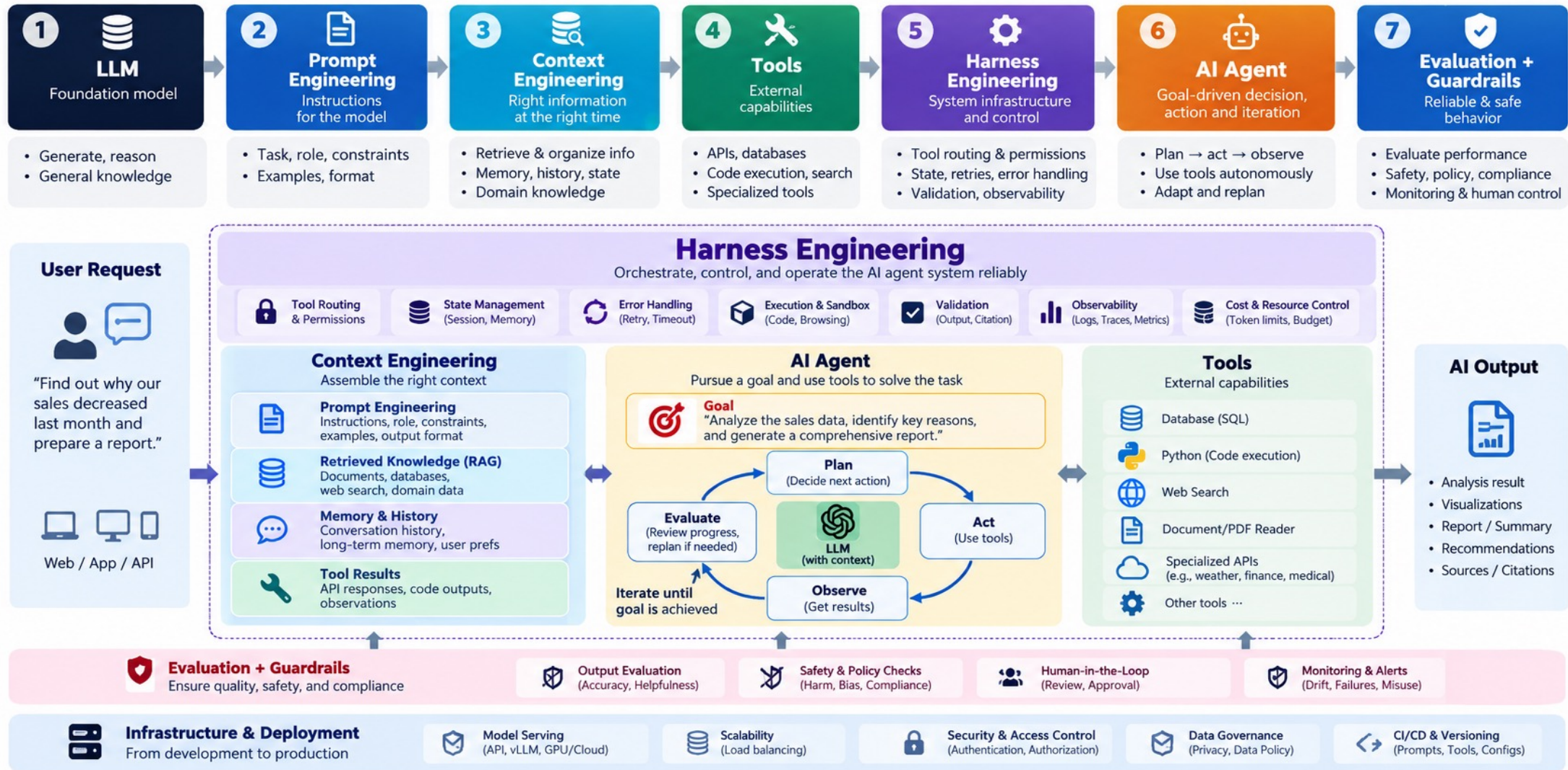
Reasoning • Computer Use • Tools • 1.05M
Context

Source: OpenAI official model documentation, Sep 2026

Note: OpenAI has not publicly disclosed parameter counts for GPT-4, GPT-5, or GPT-6.

The Big Picture: Modern GenAI System

From a foundation model to a reliable, real-world AI system





Good luck! 😊